

YEAR 7 MATHEMATICS

**Chapter 1**

# Integers

MIXED PRACTICE WORKSHEETS

# 1

## How to use this booklet

- 1 Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- 2 Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- 3 Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- 4 Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

### WE DO

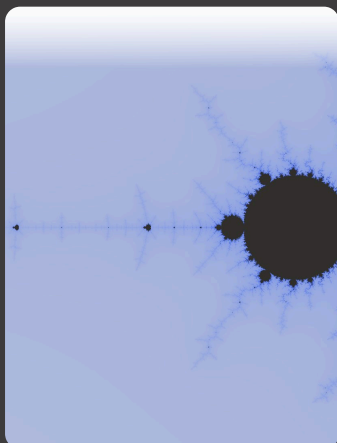
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

## Contents and progress tracker

| Lesson      | Topic                             | Page      | Date | Done                     |
|-------------|-----------------------------------|-----------|------|--------------------------|
| <b>1.01</b> | Integers and opposites            | <b>3</b>  |      | <input type="checkbox"/> |
| <b>1.02</b> | The number line                   | <b>5</b>  |      | <input type="checkbox"/> |
| <b>1.03</b> | Comparing and ordering integers   | <b>7</b>  |      | <input type="checkbox"/> |
| <b>1.04</b> | Adding integers                   | <b>9</b>  |      | <input type="checkbox"/> |
| <b>1.05</b> | Subtracting integers              | <b>11</b> |      | <input type="checkbox"/> |
| <b>1.06</b> | Multiplying integers              | <b>13</b> |      | <input type="checkbox"/> |
| <b>1.07</b> | Dividing integers                 | <b>15</b> |      | <input type="checkbox"/> |
| <b>1.08</b> | Order of operations with integers | <b>17</b> |      | <input type="checkbox"/> |
| <b>1.09</b> | Integers on a calculator          | <b>19</b> |      | <input type="checkbox"/> |
| <b>1.10</b> | Integer problems                  | <b>21</b> |      | <input type="checkbox"/> |

## In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

An integer is a whole number: positive, negative or zero. The opposite of a number is the same distance from 0, on the other side of 0.

Watch your teacher.  
Copy every step.

**E1** Write an integer for 5 m below sea level.

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**E2** What is the opposite of -6?

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**E3**  $-(-7)$

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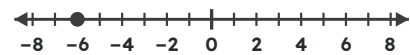
## Easy

Questions 1-9

**1** Write an integer for spending \$36.

**2** What is the opposite of -20?

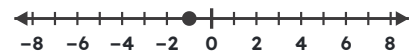
**3** Which integer is at the dot?



**4** Write an integer for winning 10 points.

**5** What is the opposite of 27?

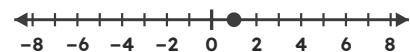
**6** Which integer is at the dot?



**7** Write an integer for spending \$18.

**8** What is the opposite of 25?

**9** Which integer is at the dot?

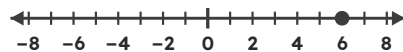


## Easy

Questions 10-18

**10** What is the opposite of 4?

**11** Which integer is at the dot?



**12** Write an integer for a drop of 10 cm.

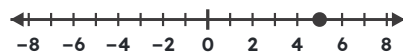
**13**  $-(-12)$

= the opposite of \_\_\_\_

= \_\_\_\_

**14** What is the opposite of 19?

**15** Which integer is at the dot?



**16** Write an integer for a deposit of \$34.

**17**  $-(-1)$

**18** What is the opposite of -27?

## Medium

Questions 19-26

**19**  $-(-17)$

**20** Which is further from 0: -14 or 6?

**21** Which two integers are 37 steps from 0?

**22** The opposite of "8 up" is "8 down". Write both as integers.

**23**  $-(+15)$

**24** Which is further from 0: -3 or 10?

**25** Which two integers are 7 steps from 0?

**26** The opposite of "12 forwards" is "12 backwards". Write both as integers.

## Medium

Questions 27–37

- 27 Which is further from 0:  $-9$  or  $10$ ?

- 28 Which two integers are 40 steps from 0?

- 29 The opposite of "11 forwards" is "11 backwards". Write both as integers.

- 30  $-(-18)$

- 31 Two opposites are 28 steps apart. What are they?

- 32 Which is further from 0:  $-9$  or  $2$ ?

- 33 Which two integers are 13 steps from 0?

- 34  $5 \times 6 - 5 \times 5$

from Year 6

- 35 The opposite of "16 up" is "16 down". Write both as integers.

- 36  $-(+16)$

- 37 Two opposites are 44 steps apart. What are they?

## Challenging

Questions 38–47

- 38 Two opposites are 34 steps apart. What are they?

- 39  $-(-(-13))$

- 40 Mia says, "The opposite of a number is always negative." Is Mia right? Explain.

- 41 Two opposites are 58 steps apart. What are they?

- 42  $-(-(-15))$

- 43 What is the opposite of 0? Explain.

- 44 Two opposites are 52 steps apart. What are they?

- 45  $100 - 39$

from Year 6

- 46  $-(-(-14))$

- 47 Ben says  $-(-4) = -4$ . Is Ben right? Explain.

## Challenging

Questions 48–57

- 48  $-(-(-10))$

- 49 Which integers from  $-6$  to  $6$  are more than 4 steps from 0?

- 50 Two opposites are 38 steps apart. What are they?

- 51 The opposite of "15 forwards" is "15 backwards". Write both as integers.

- 52  $-(-(-11))$

- 53 A number is 3 steps from 0 and less than 0. What is its opposite?

- 54 Two opposites are 36 steps apart. What are they?

- 55 Which is largest: 7 671, 6 680, 5 782, 1 812?

from Year 6

- 56 The opposite of "16 forwards" is "16 backwards". Write both as integers.

- 57  $-(-(-8))$

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

Numbers get larger to the right. To find a distance, count the gaps. If the trip passes 0, count to 0, then count on.

Watch your teacher.  
Copy every step.

E1 How far is it from -3 to 5?

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E2 Start at 4. Move 9 left. Where do you land?

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E3 Which integer is halfway between -8 and 2?

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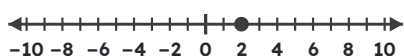
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## Easy

Questions 1-9

1 Which integer is at the dot?

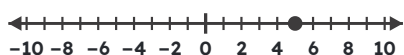


2 Start at 0. Move 2 left. Where do you land?

3 How far is it from -5 to 7?

4 Write the integers between -6 and -2.

5 Which integer is at the dot?

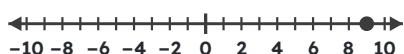


6 Start at 0. Move 7 right. Where do you land?

7 How far is it from -8 to 4?

8 Write the integers between -5 and -1.

9 Which integer is at the dot?



## Easy

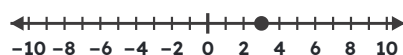
Questions 10-18

10 Start at 4. Move 9 right. Where do you land?

11 How far is it from -3 to 9?

12 Write the integers between -7 and -2.

13 Which integer is at the dot?



14 How far is it from -13 to 16?

\_\_\_ + \_\_\_ = \_\_\_

15 What is the opposite of -19?

from 1.01

16 Start at -6. Move 7 left. Where do you land?

17 How far is it from -8 to 9?

18 Write the integers between -6 and -3.

## Medium

Questions 19-26

19 How far is it from -9 to 16?

20 Which integer is halfway between -10 and 6?

21 A number line counts by 5s: -10, -5, 0, ... What are the next two numbers?

22 A lift goes from level -2 to level 7. How many floors does it go up?

23 It is 10°C. It gets 14 degrees colder. What is the temperature now?

from Year 6

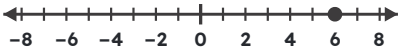
24 Point A is at -5. Point B is 14 to the left of A. Where is B?

25 How far is it from -11 to 19?

26 Which integer is halfway between -15 and 1?

## Medium

Questions 27–37

- 27 Which integer is halfway between  $-11$  and  $5$ ?
- 28 A number line counts by 5s:  $-15, -10, -5, \dots$  What are the next two numbers?
- 29 A lift goes from level  $-3$  to level  $2$ . How many floors does it go up?
- 30 Which integer is at the dot?
- 
- 31 Point A is at  $1$ . Point B is  $7$  to the left of A. Where is B?
- 32 How far is it from  $-24$  to  $19$ ?
- 33 Two integers are  $12$  apart. Halfway between them is  $-9$ . What are they?
- 34 The opposite of "15 above zero" is "15 below zero". Write both as integers.
- 35 Which integer is halfway between  $-12$  and  $-6$ ?
- 36 A number line counts by 10s:  $-20, -10, 0, \dots$  What are the next two numbers?
- 37 What is the value of the 2 in 47 269?

from 1.01

from 1.01

from 1.01

from 1.01

from Year 6

## Challenging

Questions 38–46

- 38 Two integers are  $12$  apart. Halfway between them is  $-3$ . What are they?
- 39 Start at  $2$ . Move  $6$  right, then  $11$  left. Where do you land?
- 40 How many integers are between  $-12$  and  $3$ ? Do not count  $-12$  and  $3$ .
- 41 What is the opposite of  $16$ ?
- 42 A diver is at  $-15$  m. A fish is at  $-6$  m. How far above the diver is the fish?
- 43 Two integers are  $14$  apart. Halfway between them is  $-7$ . What are they?
- 44 Start at  $1$ . Move  $5$  right, then  $15$  left. Where do you land?
- 45 Which is further from 0:  $-5$  or  $14$ ?
- 46 How many integers are between  $-4$  and  $7$ ? Do not count  $-4$  and  $7$ .

## Challenging

Questions 47–56

- 47 Start at  $5$ . Move  $7$  right, then  $9$  left. Where do you land?
- 48 How many integers are between  $-6$  and  $8$ ? Do not count  $-6$  and  $8$ .
- 49 A diver is at  $-20$  m. A fish is at  $1$  m. How far above the diver is the fish?
- 50 What is  $\frac{4}{8}$  of  $80$ ?
- 51 Two integers are  $6$  apart. Halfway between them is  $-3$ . What are they?
- 52 Point A is at  $5$ . Point B is  $9$  to the left of A. Where is B?
- 53 Start at  $-2$ . Move  $5$  right, then  $15$  left. Where do you land?
- 54 Write an integer for a loss of  $18$  kg.
- 55 How many integers are between  $-5$  and  $6$ ? Do not count  $-5$  and  $6$ .
- 56 A diver is at  $-10$  m. A fish is at  $-1$  m. How far above the diver is the fish?

from Year 6

from 1.01

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

The number further right on the number line is larger. So  $-2 > -7$ , because  $-2$  is further right. Ascending: smallest to largest.

Watch your teacher.  
Copy every step.

E1 Write  $<$  or  $>$ :  $-9$   $\square$   $-4$

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E2 Order from smallest to largest: 3,  $-5$ , 0,  $-1$

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E3 Which is colder:  $-12^{\circ}\text{C}$  or  $-8^{\circ}\text{C}$ ?

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## Easy

Questions 1–9

1 Write  $<$  or  $>$ :  $-10$   $\square$   $-6$

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2 Which is larger: 3 or  $-5$ ?

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3 Which is smaller:  $-8$  or  $-18$ ?

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4 Which is colder:  $-8^{\circ}\text{C}$  or  $-10^{\circ}\text{C}$ ?

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5 Write  $<$  or  $>$ :  $-10$   $\square$   $-7$

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6 Which is larger: 1 or  $-18$ ?

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7 Which is smaller: 2 or  $-6$ ?

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8 Which is colder:  $-7^{\circ}\text{C}$  or  $-10^{\circ}\text{C}$ ?

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9 Start at 6. Move 5 left. Where do you land?

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from 1.02

## Easy

Questions 10–18

10 Which is larger:  $-9$  or  $-15$ ?

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11 Which is smaller:  $-15$  or  $-2$ ?

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12 Which is colder:  $-2^{\circ}\text{C}$  or  $-15^{\circ}\text{C}$ ?

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13  $100 - 29$

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from Year 6

14 Write  $<$  or  $>$ : 4  $\square$   $-10$

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15 Order from smallest to largest:  $-6$ , 12, 3, 0,  $-8$

smallest: \_\_\_

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16 Which is larger:  $-10$  or 1?

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17 What is the opposite of 1?

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from 1.01

18 Which is smaller:  $-19$  or 1?

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## Medium

Questions 19–26

19 Order from smallest to largest:  $-4$ , 5, 0,  $-11$ , 10

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20 Order from largest to smallest: 6,  $-6$ ,  $-18$ ,  $-28$

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21 A number line counts by 10s:  $-30$ ,  $-20$ ,  $-10$ , ... What are the next two numbers?

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from 1.02

22 Write  $<$  or  $>$ :  $-13$   $\square$   $-44$

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23 Write all the integers between  $-9$  and  $-6$ .

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24  $42 \div 6$

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from Year 6

25 Order from smallest to largest: 0,  $-3$ , 6,  $-12$ , 8

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26 Order from largest to smallest:  $-23$ ,  $-15$ , 13,  $-4$

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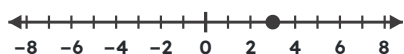
## Medium

Questions 27–37

- 27 Order from largest to smallest:  
-27, -10, -2, 9

- 28 Write  $<$  or  $>$ :  $-47 \square -20$

- 29 Which integer is at the dot?



from 1.01

- 30 Write all the integers between -8 and -5.

- 31 A lift goes from level -2 to level 9. How many floors does it go up?

from 1.02

- 32 Order from smallest to largest:  
-11, 9, -3, 6, 0

- 33 Dawn temperatures: Oberon  $8^{\circ}\text{C}$ , Cooma  $-10^{\circ}\text{C}$ , Jindabyne  $0^{\circ}\text{C}$ , Thredbo  $-3^{\circ}\text{C}$ . Which town was coldest? Which was warmest?

- 34 What is  $\frac{5}{8}$  of 80?

from Year 6

- 35 Order from largest to smallest:  
-27, -15, 5, -7

- 36 Write an integer for 38 m above sea level.

from 1.01

- 37 Write  $<$  or  $>$ :  $-19 \square -31$

## Challenging

Questions 38–47

- 38 Dawn temperatures: Oberon  $-3^{\circ}\text{C}$ , Cooma  $5^{\circ}\text{C}$ , Jindabyne  $-7^{\circ}\text{C}$ , Thredbo  $0^{\circ}\text{C}$ . Which town was coldest? Which was warmest?

- 39 True or false:  $-9 > -12$ . Explain.

- 40 Point A is at -10. Point B is 8 to the left of A. Where is B?

from 1.02

- 41 Write the smallest and the largest integer between -15 and 5.

- 42  $38 - 10 \div 5$

from Year 6

- 43 Sam says -8 is larger than -3 because 8 is larger than 3. Is Sam right? Explain.

- 44 Dawn temperatures: Oberon  $5^{\circ}\text{C}$ , Cooma  $-1^{\circ}\text{C}$ , Jindabyne  $0^{\circ}\text{C}$ , Thredbo  $-10^{\circ}\text{C}$ . Which town was coldest? Which was warmest?

- 45 What is the opposite of -6?

from 1.01

- 46 True or false:  $-6 > -10$ . Explain.

- 47 How far is it from -18 to 37?

from 1.02

## Challenging

Questions 48–57

- 48 True or false:  $-8 < -11$ . Explain.

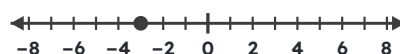
- 49 Write the smallest and the largest integer between -15 and 2.

- 50 Which is largest: 9 269, 6 177, 7 046, 6 408?

from Year 6

- 51 Order from coldest to warmest:  $2^{\circ}\text{C}$ ,  $-11^{\circ}\text{C}$ ,  $-4^{\circ}\text{C}$ ,  $0^{\circ}\text{C}$ ,  $-15^{\circ}\text{C}$

- 52 Which integer is at the dot?



from 1.01

- 53 Dawn temperatures: Oberon  $-6^{\circ}\text{C}$ , Cooma  $-3^{\circ}\text{C}$ , Jindabyne  $0^{\circ}\text{C}$ , Thredbo  $4^{\circ}\text{C}$ . Which town was coldest? Which was warmest?

- 54 Write all the integers between -4 and 0.

- 55 Which integer is halfway between -13 and 1?

from 1.02

- 56 True or false:  $-2 > -10$ . Explain.

- 57  $600 + 193$

from Year 6

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

Adding a positive moves right. Adding a negative moves left:  $+$   $(-4)$  means  $- 4$ . A  $+$  and a  $-$ : take the difference, and the sign with more wins.

Watch your teacher.  
Copy every step.

E1  $-3 + 5$

E2  $4 + (-9)$

E3  $-6 + (-2)$

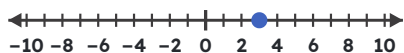
## Easy

Questions 1–12

1  $-2 + 9$

2  $3 + (-9)$

3 Use the number line to find  $3 + 5$ .

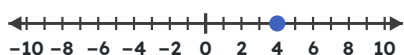


4  $-7 + (-8)$

5  $-1 + 6$

6  $9 + (-6)$

7 Use the number line to find  $4 + 3$ .



8  $-9 + (-8)$

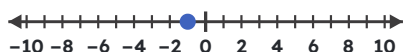
9 Start at 3. Move 4 left. Where do you land?

from 1.02

10  $-9 + 9$

11  $8 + (-2)$

12 Use the number line to find  $-1 + 3$ .

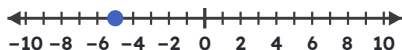


## Easy

Questions 13–21

13  $4 + (-7)$

14 Use the number line to find  $-5 + (-2)$ .



15  $-7 + (-4)$

16  $70 \div 7$

from Year 6

17  $-8 + 7$

18  $-38 + (-19)$

= \_\_\_\_ - \_\_\_\_

= \_\_\_\_

19  $1 + (-8)$

20 Which is larger:  $-17$  or  $-3$ ?

from 1.03

21 Use the number line to find  $0 + 1$ .



## Medium

Questions 22–29

22  $-19 + (-17)$

23  $15 + (-44)$

24 Which is further from 0:  $-2$  or  $13$ ?

from 1.01

25 It is  $-12^{\circ}\text{C}$ . The temperature rises 13 degrees. What is it now?

26  $-6 + 3 + (-10)$

27 What is the value of the 3 in 30 458?

from Year 6

28  $-27 + (-18)$

29  $19 + (-43)$

## Medium

Questions 30–40

30  $9 + (-32)$

31 It is  $-10^{\circ}\text{C}$ . The temperature rises 8 degrees. What is it now?

32 How far is it from  $-5$  to  $9$ ?

from 1.02

33  $-7 + 7 + (-11)$

34 Write  $<$  or  $>$ :  $-21 \square -20$ 

from 1.03

35  $-27 + (-24)$

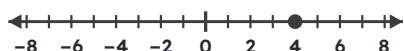
36 Find the missing number:  
 $-5 + \square = -8$ 

37  $7 \times 7 - 2 \times 4$

from Year 6

38  $27 + (-38)$

39 Which integer is at the dot?



from 1.01

40 It is  $-9^{\circ}\text{C}$ . The temperature rises 13 degrees. What is it now?

## Challenging

Questions 41–50

41 Find the missing number:  
 $-10 + \square = -6$ 

42 Kai has \$47, spends \$90, then earns \$27. What is the balance now?

43 A lift goes from level  $-1$  to level 8. How many floors does it go up?

from 1.02

44 Ruby says  $-3 + (-7) = 10$ . Is Ruby right? Explain.

45 Write 0.1 as a fraction.

from Year 6

46 A submarine at  $-100$  m rises 100 m, then sinks 70 m. How deep is it now?

47  $-11 + 5 + (-16) + 2$

48 Which is colder:  $-6^{\circ}\text{C}$  or  $-11^{\circ}\text{C}$ ?

from 1.03

49 Find the missing number:  
 $-12 + \square = -5$ 

50 The opposite of "19 up" is "19 down". Write both as integers.

from 1.01

## Challenging

Questions 51–60

51 Leo has \$54, spends \$65, then earns \$22. What is the balance now?

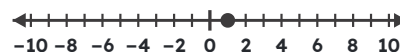
52 Tom says  $-8 + (-6) = 14$ . Is Tom right? Explain.

53 It is  $10^{\circ}\text{C}$ . It gets 15 degrees colder. What is the temperature now?

from Year 6

54 A lift is at level  $-1$ . It goes up 5 floors, then down 3. Which level is it on now?

55 Which integer is at the dot?



from 1.02

56  $-9 + 12 + (-13) + 4$

57 Find the missing number:  
 $-4 + \square = -2$ 

58 Order from smallest to largest: 0,  $-2$ ,  $-9$ , 1, 8

from 1.03

59  $4 + 2 + (-10)$

60  $10 \times 11$

from Year 6

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

Subtracting a positive moves left. Subtracting a negative moves right:  $-(-3)$  means

Watch your teacher. Copy every step.

E1  $2 - 7$

E2  $5 - (-3)$

E3  $-4 - (-9)$

## Easy

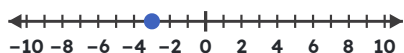
Questions 1-12

1  $1 - 7$

2  $1 - (-8)$

3  $-5 - 6$

4 Use the number line to find  $-3 - (-1)$ .

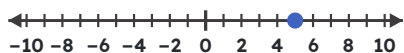


5  $2 - 6$

6  $4 - (-7)$

7  $-7 - 5$

8 Use the number line to find  $5 - (-2)$ .



9  $8 + (-3)$

from 1.04

10  $4 - 8$

11  $3 - (-2)$

12  $-6 - 5$

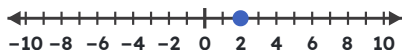
## Easy

Questions 13-22

13  $3 - (-5)$

14  $-9 - 7$

15 Use the number line to find  $2 - 5$ .



16 Write 0.1 as a fraction.

from Year 6

17  $4 - 6$

18  $-17 - (-7)$

= \_\_\_\_ + \_\_\_\_

= \_\_\_\_

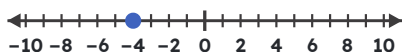
19  $3 - (-8)$

20 Which is larger: 3 or -5?

from 1.03

21  $-3 - 3$

22 Use the number line to find  $-4 - (-2)$ .



## Medium

Questions 23-29

23  $-11 - (-21)$

24 The temperature went from  $-10^{\circ}\text{C}$  to  $13^{\circ}\text{C}$ . How much did it rise?

25 Which is further from 0: -4 or 10?

from 1.01

26 Find the difference between 28 and -26.

27  $-37 - (-49)$

28 Which integer is halfway between -8 and -2?

from 1.02

29  $-14 - (-6)$

## Medium

Questions 30–39

- 30 The temperature went from  $-8^{\circ}\text{C}$  to  $4^{\circ}\text{C}$ . How much did it rise?

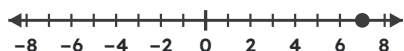
- 31 Find the difference between 12 and  $-10$ .

- 32  $22 - 24 \div 4$

from Year 6

- 33  $-22 - (-6)$

- 34 Which integer is at the dot?

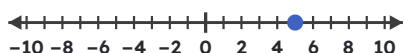


from 1.01

- 35  $-13 - (-6)$

- 36 Find the missing number:  
 $\square - (-4) = 0$

- 37 Use the number line to find  $5 + 5$ .



from 1.04

- 38 The temperature went from  $-2^{\circ}\text{C}$  to  $15^{\circ}\text{C}$ . How much did it rise?

- 39 Write  $<$  or  $>$ :  $-28$   $\square$   $-11$

from 1.03

## Challenging

Questions 40–49

- 40 Find the missing number:  
 $\square - (-2) = 1$

- 41  $2 - (-5) - 5$

- 42 Which is largest: 1 898, 6 667, 7 624, 6 433?

from Year 6

- 43 A submarine is at  $-180$  m. A plane is at 760 m. How far apart are they?

- 44 How far is it from  $-8$  to 3?

from 1.02

- 45 Leo says  $6 - (-4) = 2$ . Is Leo right? Explain.

- 46 Find the missing number:  
 $\square - (-3) = 5$

- 47 The opposite of "5 up" is "5 down". Write both as integers.

from 1.01

- 48  $-2 - (-6) - 10$

- 49  $-7 + 5 + (-7)$

from 1.04

## Challenging

Questions 50–59

- 50  $-4 - (-4) - 5$

- 51 A submarine is at  $-190$  m. A plane is at 320 m. How far apart are they?

- 52  $121 \div 11$

from Year 6

- 53 Aisha says  $5 - (-7) = -2$ . Is Aisha right? Explain.

- 54 What is the opposite of  $-15$ ?

from 1.01

- 55 Find the missing number:  
 $\square - (-8) = 9$

- 56  $-35 - (-23)$

- 57 Which is colder:  $-4^{\circ}\text{C}$  or  $-19^{\circ}\text{C}$ ?

from 1.03

- 58  $-2 - (-4) - 14$

- 59 A lift goes from level  $-1$  to level 6. How many floors does it go up?

from 1.02

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

Multiply the numbers, then find the sign. Same signs: the answer is positive. Different signs: the answer is negative.

Watch your teacher.  
Copy every step.

E1  $-4 \times 3$

E2  $-5 \times (-6)$

E3  $-2 \times 3 \times (-4)$

## Easy

Questions 1–12

1  $-5 \times 3$

2  $-2 \times (-3)$

3 Positive or negative? Do not work it out:  $7 \times 11$

4 Write  $5 \times (-4)$  as a sum, then find the answer.

= \_\_\_\_

5  $3 \times (-4)$

6  $-9 \times (-9)$

7 Positive or negative? Do not work it out:  $4 \times 7$

8 Write  $3 \times (-3)$  as a sum, then find the answer.

9  $9 + (-7)$

from 1.04

10  $-3 \times 3$

11  $-8 \times (-6)$

12 Positive or negative? Do not work it out:  $-8 \times 12$

## Easy

Questions 13–21

13  $-8 \times (-9)$

14 Positive or negative? Do not work it out:  $4 \times 12$

15 Write  $2 \times (-2)$  as a sum, then find the answer.

(\_\_\_\_) + (\_\_\_\_)

= \_\_\_\_

16  $28 - 12 \div 4$

from Year 6

17  $-8 \times 8$

18  $-15 \times (-3)$

19  $-5 \times (-8)$

20  $7 - (-5)$

from 1.05

21 Positive or negative? Do not work it out:  $5 \times 11$

## Medium

Questions 22–29

22  $-21 \times (-6)$

23  $1 \times (-1) \times 5$

24 Which is further from 0:  $-4$  or 11?

from 1.01

25  $(-12)^2$

26 The temperature falls  $3^\circ\text{C}$  each hour for 5 hours. Write a multiplication for the change.

27 Order from largest to smallest: 3,  $-21$ ,  $-10$ ,  $-3$

from 1.03

28  $-18 \times (-4)$

29  $-4 \times (-3) \times (-5)$

## Medium

Questions 30–40

30  $2 \times (-4) \times (-3)$

31  $(-3)^2$

32 Write 0.9 as a fraction.

from Year 6

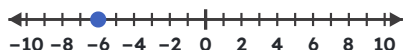
33 The temperature falls  $5^\circ\text{C}$  each hour for 3 hours. Write a multiplication for the change.

34 Start at  $-3$ . Move 5 right. Where do you land?

from 1.02

35  $-21 \times (-4)$

36 Find the missing number:  
 $\square \times (-7) = -49$ 

37 Use the number line to find  $-6 + 1$ .


from 1.04

38  $-2 \times 1 \times 2$

39 Find the difference between 26 and  $-10$ .

from 1.05

40  $(-2)^2$

## Challenging

Questions 41–50

41 Find the missing number:  
 $\square \times 2 = 16$ 

42 Find two integers that multiply to  $-56$  and add to  $-1$ .

43 It is  $8^\circ\text{C}$ . It gets 16 degrees colder. What is the temperature now?

from Year 6

44  $-1 \times (-1) \times (-1) \times (-1) \times (-1) \times (-1)$

45 Which is smaller:  $-12$  or  $-2$ ?

from 1.03

46 Ruby says  $(-8)^2 = -64$ . Is Ruby right? Explain.

47 Find the missing number:  
 $\square \times (-5) = 30$ 

48 Which two integers are 19 steps from 0?

from 1.01

49 Find two integers that multiply to  $-48$  and add to 2.

50  $-6 + 3 + (-8)$

from 1.04

## Challenging

Questions 51–60

51 Find two integers that multiply to  $-40$  and add to 3.

52  $-1 \times (-1) \times (-1) \times (-1) \times (-1) \times (-1) \times (-1) \times (-1)$

53 What is the value of the 1 in 83 175?

from Year 6

54 Tom says  $(-8)^2 = -64$ . Is Tom right? Explain.

55 How far is it from  $-2$  to 5?

from 1.02

56 Find the missing number:  
 $\square \times (-7) = 63$ 

57 The temperature falls  $6^\circ\text{C}$  each hour for 6 hours. Write a multiplication for the change.

58 Use the number line to find  $-1 - 1$ .


from 1.05

59 Find two integers that multiply to  $-8$  and add to 2.

60 Write all the integers between  $-8$  and  $-4$ .

from 1.03

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

Divide the numbers, then find the sign. The rules are the same as for  $\times$ : same signs give  $+$ , different signs give  $-$ .

Watch your teacher.  
Copy every step.

E1  $-20 \div 4$

E2  $-36 \div (-9)$

E3 Find the average:  $-3, 5, -8, -6$

## Easy

Questions 1–12

1  $-24 \div 3$

2  $-18 \div (-2)$

3  $45 \div (-5)$

4 Positive or negative? Do not work it out:  $6 \div (-6)$

5  $-36 \div 4$

6  $-21 \div (-7)$

7  $12 \div (-6)$

8 Positive or negative? Do not work it out:  $-24 \div 3$

9  $-7 \times (-7)$

from 1.06

10  $-8 \div 2$

11  $-18 \div (-9)$

12  $28 \div (-7)$

## Easy

Questions 13–22

13  $-28 \div (-4)$

14  $49 \div (-7)$

15 Positive or negative? Do not work it out:  $-24 \div (-8)$

16 Write 0.3 as a fraction.

from Year 6

17  $-49 \div 7$

18  $-77 \div (-7)$

same signs: \_\_\_\_\_

= \_\_\_\_

19  $-8 \div (-2)$

20  $-1 - (-4)$

from 1.05

21  $18 \div (-9)$

22 Positive or negative? Do not work it out:  $-12 \div (-3)$

## Medium

Questions 23–29

23  $-72 \div 6$

24 Find the average of these temperatures:  $-3^\circ\text{C}, 5^\circ\text{C}, 4^\circ\text{C}, -5^\circ\text{C}, -6^\circ\text{C}$

25 Which is further from 0:  $-11$  or  $14$ ?

from 1.01

26 A diver goes down 42 m in 7 equal stages. Write a division for each stage.

27 Work out  $\frac{-12}{-6}$

28  $22 + (-15)$

from 1.04

29  $-88 \div (-8)$

## Medium

Questions 30–39

- 30 Find the average of these temperatures:  $-1^{\circ}\text{C}$ ,  $6^{\circ}\text{C}$ ,  $-8^{\circ}\text{C}$ ,  $-1^{\circ}\text{C}$
- 31 A diver goes down 35 m in 7 equal stages. Write a division for each stage.
- 32  $22 - 20 \div 5$
- 33 Work out  $\frac{-81}{9}$
- 34 Start at 1. Move 5 left. Where do you land?
- 35  $-55 \div (-5)$
- 36 Find the missing number:  
 $\square \div 7 = 1$
- 37 Positive or negative? Do not work it out:  $-11 \times 5$
- 38 Find the average of these temperatures:  $-11^{\circ}\text{C}$ ,  $3^{\circ}\text{C}$ ,  $3^{\circ}\text{C}$ ,  $5^{\circ}\text{C}$
- 39 Find the difference between 21 and  $-12$ .

from Year 6

from 1.02

from 1.06

from 1.05

## Challenging

Questions 40–49

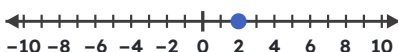
- 40 Find the missing number:  
 $\square \div 3 = -12$
- 41 Find the missing number:  
 $-12 \div \square = 4$
- 42 It is  $9^{\circ}\text{C}$ . It gets 14 degrees colder. What is the temperature now?
- 43  $-30 \div 5 \times (-2)$
- 44 Use the number line to find  $2 + (-3)$ .
- 45 Lena says  $-49 \div (-7) = -7$ . Is Lena right? Explain.
- 46 Find the missing number:  
 $\square \div 3 = 5$
- 47 Order from largest to smallest:  $-23$ ,  $3$ ,  $-13$ ,  $-7$
- 48 Find the missing number:  
 $-14 \div \square = 7$
- 49 The temperature falls  $5^{\circ}\text{C}$  each hour for 4 hours. Write a multiplication for the change.

from Year 6

from 1.04

from 1.03

from 1.06



## Challenging

Questions 50–59

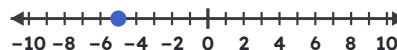
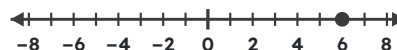
- 50 Find the missing number:  
 $18 \div \square = -9$
- 51  $4 \div (-1) \times (-4)$
- 52  $88 \div 11$
- 53 Leo says  $-18 \div (-3) = -6$ . Is Leo right? Explain.
- 54 Which integer is at the dot?
- 55 Find the missing number:  
 $\square \div (-1) = 8$
- 56 Work out  $\frac{-81}{-9}$
- 57 Use the number line to find  $-5 - (-5)$ .
- 58 Find the missing number:  
 $-6 \div \square = 2$
- 59  $1 + 9 + (-7)$

from Year 6

from 1.01

from 1.05

from 1.04



NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

**WE DO**

**KEY IDEA**

Brackets first. Then  $\times$  and  $\div$ , from left to right. Then  $+$  and  $-$ , from left to right. Write one step a line.

Watch your teacher.  
Copy every step.

E1  $-3 + 4 \times (-2)$

E2  $(-6 + 2) \times 5$

E3  $-20 \div (3 - 8)$

**Easy**

Questions 1–9

1  $-4 + 3 \times (-1)$

$= \_\_\_ + (\_\_\_)$

$= \_\_\_$

2  $4 \times (-3) - 5$

3  $(4 + 1) \times (-4)$

4  $9 + 3 \times 5$

5  $5 \times 6 - 10$

6  $(9 + (-7)) \times 4$

7  $7 + 4 \times 2$

8  $6 \times (-3) - 10$

9  $-9 \times (-4)$

from 1.06

**Easy**

Questions 10–18

10  $5 \times 2 - 12$

$= \_\_\_ - \_\_\_$

$= \_\_\_$

11  $(-8 + (-1)) \times 2$

12  $8 + 6 \times 4$

13  $24 - 18 \div 6$

from Year 6

14  $-8 \div 2 + (-5) \times 5$

15  $5 \times (-3) - 5$

16  $(-1 + 4) \times 1$

17  $-48 \div (-6)$

from 1.07

18  $1 + 3 \times 1$

**Medium**

Questions 19–26

19  $18 \div 3 + (-4) \times 3$

20  $-7 - (6 - (-7))$

21 Which is further from 0:  $-15$  or 5?

from 1.01

22  $(-6)^2 - 30$

23  $5 \div (1 - 6)$

24 The temperature went from  $-5^\circ\text{C}$  to  $9^\circ\text{C}$ . How much did it rise?

from 1.05

25  $-20 \div (-5) + 1 \times 5$

26  $-11 - (9 - (-2))$

**Medium**

Questions 27–37

27  $-8 - (4 - (-8))$

28  $(-3)^2 - 13$

29 What is the value of the 9 in 96 406?

from Year 6

30  $-6 \div (3 - 6)$

 31 Start at  $-2$ . Move 3 left. Where do you land?

from 1.02

32  $2 \div 2 + (-5) \times 2$

33  $(-8 - (-9)) \times (1 + 6)$

 34 Positive or negative? Do not work it out:  $-2 \times 12$ 

from 1.06

35  $2 - (1 - 3)$

36 A diver goes down 96 m in 8 equal stages. Write a division for each stage.

from 1.07

37  $(-4)^2 - 19$

**Challenging**

Questions 38–47

38  $(6 - (-5)) \times (-5 + (-2))$

39  $[-7 + 2 \times 2] \div 3$

40 100 students sit in 4 equal rows. How many are in each row?

from Year 6

 41 Put brackets in to make this true:  $-5 + 3 \times 3 = -6$ 

42  $-4 - 5$

from 1.05

 43 Zac says  $7 - 2 \times (-4) = -20$ . Is Zac right? Explain.

44  $(2 - 6) \times (6 + 6)$

 45 Order from largest to smallest:  $-26, -16, 1, -5$ 

from 1.03

46  $[-8 + 3 \times (-4)] \div 4$

 47 The temperature falls  $5^\circ\text{C}$  each hour for 8 hours. Write a multiplication for the change.

from 1.06

**Challenging**

Questions 48–57

48  $[-3 + 5 \times (-3)] \div 3$

 49 Put brackets in to make this true:  $-9 + 5 \times 2 = -8$ 

50 Write 0.7 as a fraction.

from Year 6

 51 Leo says  $7 - 3 \times (-5) = -20$ . Is Leo right? Explain.

52  $6 + (-2)$

from 1.04

53  $(-6 - 9) \times (6 + (-1))$

54  $-15 \div (0 - 3)$

 55 Positive or negative? Do not work it out:  $10 \div 2$ 

from 1.07

56  $[-5 + 5 \times (-3)] \div 5$

57  $-35 - (-14)$

from 1.05

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

## WE DO

## KEY IDEA

Use the  $(-)$  key for a negative number and the  $-$  key to subtract. Estimate first, then check that the calculator answer is close.

Watch your teacher.  
Copy every step.

**E1** Which keys do you press for  $-8 \times 5$ ?

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**E2** Use a calculator:  $-348 + 129$

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**E3** Estimate, then use a calculator:  $-394 \times 21$

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## Easy

Questions 1–9

**1** Use a calculator:  $-521 + 189$

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**2** Use a calculator:  $-19 \times 11$

---

**3** Which keys do you press for  $-7 + 2$ ?

---

**4** Use a calculator:  $-957 \div (-29)$

---

**5** Use a calculator:  $-433 + 553$

---

**6** Use a calculator:  $-40 \times 22$

---

**7** Which keys do you press for  $-6 \times 4$ ?

---

**8** Use a calculator:  $-506 \div (-23)$

---

**9**  $2 \times 1 - 7$

from 1.08

## Easy

Questions 10–18

**10** Use a calculator:  $-30 \times 27$

---

**11** Which keys do you press for  $-5 - 8$ ?

---

**12** Use a calculator:  $-627 \div (-19)$

---

**13**  $88 \div 8$

from Year 6

**14** Use a calculator:  $-589 + 661$

---

**15** Estimate, then use a calculator:  $-42 \times 29$

$\approx -40 \times 30 =$  \_\_\_\_\_

$=$  \_\_\_\_\_

**16** Use a calculator:  $-46 \times 38$

---

**17**  $-63 \div (-9)$

from 1.07

**18** Which keys do you press for  $-6 + 4$ ?

---

## Medium

Questions 19–26

**19** Estimate, then use a calculator:  $-49 \times 29$

---

**20** Use a calculator:  $(-23)^2$

---

**21** Which is further from 0:  $-13$  or  $9$ ?

from 1.01

**22** Use a calculator:  $(-281 + 53) \times (-11)$

---

**23** Use a calculator:  $-1\,295 \div 37 - 296$

---

**24**  $-5 \times (-1) \times 6$

from 1.06

**25** Estimate, then use a calculator:  $-52 \times 22$

---

**26** Use a calculator:  $(-20)^2$

---

## Medium

Questions 27–36

- 27 Use a calculator:  $(-17)^2$

- 28 Use a calculator:  
 $(-121 + 50) \times (-14)$

- 29  $39 - 30 \div 6$

from Year 6

- 30 Use a calculator:  
 $-1\,254 \div 38 - 282$

- 31 Start at 0. Move 2 right.  
Where do you land?

from 1.02

- 32 Estimate, then use a  
calculator:  $-45 \times 44$

- 33 Chloe pressed  $(-)$  8  $\times^2$  and got  
 $-64$ . What keys give  $(-8)^2$ ?  
What is the answer?

- 34  $(1 + (-5)) \times (-2)$

from 1.08

- 35 Use a calculator:  $(-12)^2$

- 36 A diver goes down 30 m in 3  
equal stages. Write a division  
for each stage.

from 1.07

## Challenging

Questions 37–46

- 37 Lena pressed  $(-)$  7  $\times^2$  and got  
 $-49$ . What keys give  $(-7)^2$ ?  
What is the answer?

- 38 Use a calculator:  
 $(-330 + -12) \div (-19)$

- 39 What is  $\frac{6}{8}$  of 80?

from Year 6

- 40 The temperature on Mars  
changed by  $-15^\circ\text{C}$  each hour  
for 14 hours. Find the  
total change.

- 41 Positive or negative? Do not  
work it out:  $10 \times 5$

from 1.06

- 42 Jack says  $-201 \times 67$  is about  
 $-140\,000$ . Is that a sensible  
estimate? Explain.

- 43 Ali pressed  $(-)$  4  $\times^2$  and got  
 $-16$ . What keys give  $(-4)^2$ ?  
What is the answer?

- 44 Order from largest to  
smallest:  $-9, 13, -26, -14$

from 1.03

- 45 Use a calculator:  
 $(-346 + 458) \div (-14)$

- 46  $2 \div (7 - 8)$

from 1.08

## Challenging

Questions 47–56

- 47 Use a calculator:  
 $(-250 + -26) \div (-12)$

- 48 The temperature on Mars  
changed by  $-34^\circ\text{C}$  each hour  
for 13 hours. Find the  
total change.

- 49  $100 - 36$

from Year 6

- 50 Aisha says  $-293 \times 73$  is about  
 $-210\,000$ . Is that a sensible  
estimate? Explain.

- 51  $6 + (-7)$

from 1.04

- 52 Zac pressed  $(-)$  4  $\times^2$  and got  
 $-16$ . What keys give  $(-4)^2$ ?  
What is the answer?

- 53 Use a calculator:  
 $-400 \div 16 - 288$

- 54 Positive or negative? Do not  
work it out:  $108 \div 9$

from 1.07

- 55 Use a calculator:  
 $(-421 + 943) \div (-18)$

- 56 The temperature falls  $2^\circ\text{C}$   
each hour for 4 hours. Write  
a multiplication for  
the change.

from 1.06

NAME \_\_\_\_\_ CLASS \_\_\_\_\_ DATE \_\_\_\_\_

**WE DO**

Watch your teacher.  
Copy every step.

**KEY IDEA**

Up, rises, gains and money in are positive. Down, falls, losses and money out are negative. Write a number sentence, then answer the question.

**E1** It is  $-4^{\circ}\text{C}$ . The temperature rises  $9^{\circ}\text{C}$ . What is it now?

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**E2** A diver at  $-12\text{ m}$  goes down  $7\text{ m}$ , then up  $15\text{ m}$ . Where is the diver now?

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**E3** The temperature falls  $3^{\circ}\text{C}$  each hour for 5 hours. What is the change?

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**Easy**

Questions 1–7

**1** It is  $0^{\circ}\text{C}$ . The temperature falls  $13^{\circ}\text{C}$ . What is it now?

\_\_\_\_ - \_\_\_\_  
= \_\_\_\_  $^{\circ}\text{C}$

**2** Priya has \$47 in the bank and spends \$68. What is the balance?

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**3** A lift starts at level  $-4$  and goes up 12 floors. Which level is it on now?

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**4** It is  $3^{\circ}\text{C}$ . The temperature falls  $4^{\circ}\text{C}$ . What is it now?

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**5** Leo has \$77 in the bank and spends \$119. What is the balance?

---

---

**6** A lift starts at level  $-4$  and goes up 5 floors. Which level is it on now?

---

---

**7** It is  $6^{\circ}\text{C}$ . The temperature falls  $5^{\circ}\text{C}$ . What is it now?

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**Easy**

Questions 8–14

**8** Mia has \$42 in the bank and spends \$56. What is the balance?

\_\_\_\_ - \_\_\_\_  
= \_\_\_\_ dollars

**9** A lift starts at level  $-1$  and goes up 8 floors. Which level is it on now?

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**10** It is  $4^{\circ}\text{C}$ . The temperature falls  $7^{\circ}\text{C}$ . What is it now?

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**11**  $90 \div 9$

from Year 6

**12** A diver at  $-5\text{ m}$  goes down  $8\text{ m}$ , then up  $20\text{ m}$ . Where is the diver now?

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**13** Lena has \$34 in the bank and spends \$64. What is the balance?

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**14** A lift starts at level  $-1$  and goes up 11 floors. Which level is it on now?

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**Medium**

Questions 15–20

**15** A diver at  $-20\text{ m}$  goes down  $7\text{ m}$ , then up  $8\text{ m}$ . Where is the diver now?

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**16** The highest temperature was  $9^{\circ}\text{C}$ . The lowest was  $-11^{\circ}\text{C}$ . What is the difference?

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**17** Which is further from 0:  $-8$  or 9?

from 1.01

**18** A tank's water level drops  $6\text{ cm}$  each day for 5 days. What is the total change?

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**19** In three rounds of golf, Tom scored  $-2, 4, 0$ . What is the total score?

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**20** Use a calculator:  $(-12)^2$

from 1.09

## Medium

Questions 21–28

- 21 The highest temperature was  $4^{\circ}\text{C}$ . The lowest was  $-3^{\circ}\text{C}$ . What is the difference?

- 22 A tank's water level drops 5 cm each day for 5 days. What is the total change?

- 23  $28 - 24 \div 4$

from Year 6

- 24 In three rounds of golf, Lena scored  $-3$ ,  $1$ ,  $-2$ . What is the total score?

- 25 Start at 6. Move 8 right. Where do you land?

from 1.02

- 26 A diver at  $-13$  m goes down 8 m, then up 25 m. Where is the diver now?

- 27 It is  $6^{\circ}\text{C}$  at 6 pm. The temperature falls  $2^{\circ}\text{C}$  every hour. What is it 5 hours later?

- 28  $5 \times 2 - 6$

from 1.08

## Challenging

Questions 29–36

- 29 It is  $12^{\circ}\text{C}$  at 6 pm. The temperature falls  $4^{\circ}\text{C}$  every hour. What is it 8 hours later?

- 30 Overnight temperatures for four nights:  $-1^{\circ}\text{C}$ ,  $-3^{\circ}\text{C}$ ,  $-15^{\circ}\text{C}$ ,  $3^{\circ}\text{C}$ . Find the average.

- 31 Which is largest: 6 881, 2 720, 5 329, 3 088?

from Year 6

- 32 Ivy owes \$51 and pays back \$25 a week for 4 weeks. What is the balance now?

- 33 Use the number line to find  $-2 + (-3)$ .



from 1.04

- 34 A plane is at 277 m. A submarine is at  $-110$  m. How much higher is the plane?

- 35 It is  $10^{\circ}\text{C}$  at 6 pm. The temperature falls  $4^{\circ}\text{C}$  every hour. What is it 7 hours later?

- 36 Order from largest to smallest:  $-17$ ,  $-8$ ,  $14$ ,  $-28$

from 1.03

## Challenging

Questions 37–45

- 37 Overnight temperatures for four nights:  $1^{\circ}\text{C}$ ,  $4^{\circ}\text{C}$ ,  $-8^{\circ}\text{C}$ ,  $-9^{\circ}\text{C}$ . Find the average.

- 38 Ali owes \$73 and pays back \$17 a week for 3 weeks. What is the balance now?

- 39  $1\,000 - 226$

from Year 6

- 40 A plane is at 401 m. A submarine is at  $-229$  m. How much higher is the plane?

- 41  $-4 \times (-3)$

from 1.06

- 42 It is  $11^{\circ}\text{C}$  at 6 pm. The temperature falls  $2^{\circ}\text{C}$  every hour. What is it 8 hours later?

- 43 In three rounds of golf, Lena scored  $-6$ ,  $4$ ,  $2$ . What is the total score?

- 44 Which keys do you press for  $-6 - 6$ ?

from 1.09

- 45 Overnight temperatures for four nights:  $-3^{\circ}\text{C}$ ,  $-10^{\circ}\text{C}$ ,  $-13^{\circ}\text{C}$ ,  $-10^{\circ}\text{C}$ . Find the average.